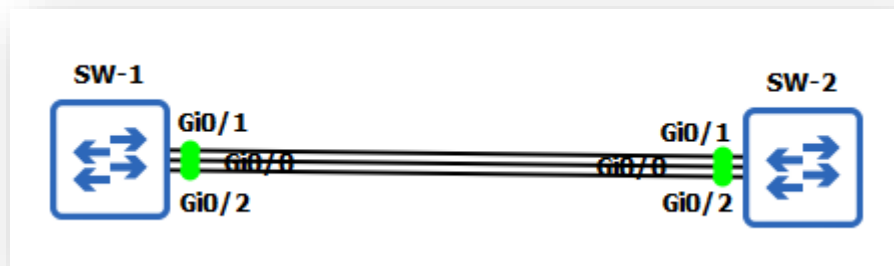


Layer-2 EtherChannel

**Task**

1. Configure the SW-1 with IP 150.150.150.1 and SW-2 with IP 150.150.150.2 for device management and hostnames.
2. Configure EtherChannel (LACP) with Group number 10
3. Configure trunk link on Port-Channel 10
4. Shut down the int Gi0/0 (member of EtherChannel) on SW-1

Task-1: Configure Hostname on switches as SW-1 and SW-2. Configure the SW-1 with IP 150.150.150.1 and SW-2 with IP 150.150.150.2 for device management.

SW-1 Configuration:

```
Switch#config t
Switch(config)#hostname SW-1
SW-1(config)#no ip domain-lookup
SW-1(config)#int vlan1
SW-1(config-if)#ip address 150.150.150.1 255.255.255.0
SW-1(config-if)#no shut
SW-1(config-if)#exit
SW-1(config)#exit
SW-1#
```

SW-2 Configuration:

```
Switch#config t
Switch(config)#hostname SW-2
SW-2(config)#no ip domain-lookup
SW-2(config)#int vlan1
SW-2(config-if)#ip address 150.150.150.2 255.255.255.0
SW-2(config-if)#no shut
SW-2(config-if)#exit
SW-2(config)#exit
SW-2#
```

✓ Verification & Testing:

SW-1#sh spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769
 Address 0c4d.da1e.4800
 Cost 4
 Port 1 (GigabitEthernet0/0)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 0c4d.daf9.9600
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Root	FWD	4	128.1	P2p	
Gi0/1	Altn	BLK	4	128.2	P2p	
Gi0/2	Altn	BLK	4	128.3	P2p	
Gi0/3	Desg	FWD	4	128.4	P2p	
Gi1/0	Desg	FWD	4	128.5	P2p	
Gi1/1	Desg	FWD	4	128.6	P2p	
Gi1/2	Desg	FWD	4	128.7	P2p	
Gi1/3	Desg	FWD	4	128.8	P2p	
Gi2/0	Desg	FWD	4	128.9	P2p	
Gi2/1	Desg	FWD	4	128.10	P2p	
Gi2/2	Desg	FWD	4	128.11	P2p	
Gi2/3	Desg	FWD	4	128.12	P2p	
Gi3/0	Desg	FWD	4	128.13	P2p	
Gi3/1	Desg	FWD	4	128.14	P2p	
Gi3/2	Desg	FWD	4	128.15	P2p	
Gi3/3	Desg	FWD	4	128.16	P2p	

SW-2#sh spanning-tree

VLAN0001

Spanning tree enabled protocol ieee
 Root ID Priority 32769
 Address 0c4d.da1e.4800
 This bridge is the root
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 0c4d.da1e.4800
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Desg	FWD	4	128.1	P2p	
Gi0/1	Desg	FWD	4	128.2	P2p	
Gi0/2	Desg	FWD	4	128.3	P2p	
Gi0/3	Desg	FWD	4	128.4	P2p	
Gi1/0	Desg	FWD	4	128.5	P2p	

Gi1/1	Desg FWD 4	128.6	P2p
Gi1/2	Desg FWD 4	128.7	P2p
Gi1/3	Desg FWD 4	128.8	P2p
Gi2/0	Desg FWD 4	128.9	P2p
Gi2/1	Desg FWD 4	128.10	P2p
Gi2/2	Desg FWD 4	128.11	P2p
Gi2/3	Desg FWD 4	128.12	P2p
Gi3/0	Desg FWD 4	128.13	P2p
Gi3/1	Desg FWD 4	128.14	P2p
Gi3/2	Desg FWD 4	128.15	P2p
Gi3/3	Desg FWD 4	128.16	P2p

Task-2: Configure EtherChannel (LACP) with Group number 10

LACP Modes:

Mode	Description
active	Places a port into an active negotiating state in which the port starts negotiations with other ports by sending LACP packets.
passive	Places a port into a passive negotiating state in which the port responds to LACP packets that it receives, but does not start LACP packet negotiation. This setting minimizes the transmission of LACP packets.

- A port in the **active** mode can form an EtherChannel with another port that is in the **active** or **passive** mode.
- A port in the **passive** mode cannot form an EtherChannel with another port that is also in the **passive** mode because neither port starts LACP negotiation.

SW-1 Configuration:

```
SW-1#config t
SW-1(config)#interface range gi0/0 - 2
SW-1(config-if-range)#channel-group 10 mode active
Creating a port-channel interface Port-channel 10
SW-1(config-if-range)#exit
SW-1(config)#exit
SW-1#
```

***Dec 16 13:30:23.510: %EC-5-L3DONTBN DL2: Gi0/2 suspended: LACP currently not enabled on the remote port.**

***Dec 16 13:30:23.634: %EC-5-L3DONTBN DL2: Gi0/1 suspended: LACP currently not enabled on the remote port.**

***Dec 16 13:30:24.051: %EC-5-L3DONTBN DL2: Gi0/0 suspended: LACP currently not enabled on the remote port.**

SW-2 Configuration:

```
SW-2#config t
SW-2(config)#interface range gi0/0 - 2
SW-2(config-if-range)#channel-group 10 mode active
Creating a port-channel interface Port-channel 10
SW-2(config-if-range)#exit
SW-2(config)#exit
SW-2#
```

```
*Dec 16 13:33:17.699: %LINK-3-UPDOWN: Interface Port-channel10, changed state to up
*Dec 16 13:33:18.699: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel10,
changed state to up
```

Task-3: Configure trunk link on Port-Channel 10**SW-1 Configuration:**

```
SW-1#config t
SW-1(config)#int port-channel 10
SW-1(config-if)#switchport trunk encapsulation dot1q
SW-1(config-if)#switchport mode trunk
SW-1(config-if)#exit
SW-1(config)#exit
SW-1#
```

SW-2 Configuration:

```
SW-2#config t
SW-2(config)#int port-channel 10
SW-2(config-if)#switchport trunk encapsulation dot1q
SW-2(config-if)#switchport mode trunk
SW-2(config-if)#exit
SW-2(config)#exit
SW-2#
```

✓ Verification & Testing:**SW-1#sh interface trunk**

Port	Mode	Encapsulation	Status	Native vlan
Po10	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Po10	1-4094

Port	Vlans allowed and active in management domain
Po10	1

Port	Vlans in spanning tree forwarding state and not pruned
Po10	1

SW-2#sh interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Po10	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Po10	1-4094

Port	Vlans allowed and active in management domain
Po10	1

Port	Vlans in spanning tree forwarding state and not pruned
Po10	1

SW-1#sh spanning-tree**VLAN0001**

Spanning tree enabled protocol ieee
 Root ID Priority 32769
 Address 0c4d.da1e.4800
 Cost 3
 Port 65 (Port-channel10)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 0c4d.daf9.9600
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/3	Desg	FWD	4	128.4	P2p
Gi1/0	Desg	FWD	4	128.5	P2p
Gi1/1	Desg	FWD	4	128.6	P2p
Gi1/2	Desg	FWD	4	128.7	P2p
Gi1/3	Desg	FWD	4	128.8	P2p
Gi2/0	Desg	FWD	4	128.9	P2p
Gi2/1	Desg	FWD	4	128.10	P2p

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi2/2	Desg	FWD	4	128.11	P2p
Gi2/3	Desg	FWD	4	128.12	P2p
Gi3/0	Desg	FWD	4	128.13	P2p
Gi3/1	Desg	FWD	4	128.14	P2p
Gi3/2	Desg	FWD	4	128.15	P2p
Gi3/3	Desg	FWD	4	128.16	P2p
Po10	Root	FWD	3	128.65	P2p

SW-2#sh spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 0c4d.da1e.4800

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c4d.da1e.4800

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/3	Desg	FWD	4	128.4	P2p	
Gi1/0	Desg	FWD	4	128.5	P2p	
Gi1/1	Desg	FWD	4	128.6	P2p	
Gi1/2	Desg	FWD	4	128.7	P2p	
Gi1/3	Desg	FWD	4	128.8	P2p	
Gi2/0	Desg	FWD	4	128.9	P2p	
Gi2/1	Desg	FWD	4	128.10	P2p	
Gi2/2	Desg	FWD	4	128.11	P2p	

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi2/3	Desg	FWD	4	128.12	P2p	
Gi3/0	Desg	FWD	4	128.13	P2p	
Gi3/1	Desg	FWD	4	128.14	P2p	
Gi3/2	Desg	FWD	4	128.15	P2p	
Gi3/3	Desg	FWD	4	128.16	P2p	
Po10	Desg	FWD	3	128.65	P2p	

SW-1#sh int status | exclude notconnect

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	a-full	auto	RJ45
Gi0/1		connected	trunk	a-full	auto	RJ45
Gi0/2		connected	trunk	a-full	auto	RJ45
Po10		connected	trunk	a-full	auto	

SW-2#sh int status | exclude notconnect

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	a-full	auto	RJ45
Gi0/1		connected	trunk	a-full	auto	RJ45
Gi0/2		connected	trunk	a-full	auto	RJ45
Po10		connected	trunk	a-full	auto	

SW-1#sh lacp neighbor

Flags: S - Device is requesting Slow LACPDUs
 F - Device is requesting Fast LACPDUs
 A - Device is in Active mode P - Device is in Passive mode

Channel group 10 neighbors

Partner's information:

LACP port					Admin	Oper	Port	Port
Port	Flags	Priority	Dev ID	Age	key	Key	Number	State
Gi0/0	SA	32768	0c4d.da1e.8000	9s	0x0	0xA	0x1	0x3D
Gi0/1	SA	32768	0c4d.da1e.8000	10s	0x0	0xA	0x2	0x3D
Gi0/2	SA	32768	0c4d.da1e.8000	8s	0x0	0xA	0x3	0x3D

SW-2#sh lacp neighbor

Flags: S - Device is requesting Slow LACPDUs
 F - Device is requesting Fast LACPDUs
 A - Device is in Active mode P - Device is in Passive mode

Channel group 10 neighbors

Partner's information:

LACP port					Admin	Oper	Port	Port
Port	Flags	Priority	Dev ID	Age	key	Key	Number	State
Gi0/0	SA	32768	0c4d.daf9.8000	6s	0x0	0xA	0x1	0x3D
Gi0/1	SA	32768	0c4d.daf9.8000	5s	0x0	0xA	0x2	0x3D
Gi0/2	SA	32768	0c4d.daf9.8000	10s	0x0	0xA	0x3	0x3D

SW-1#sh etherchannel protocol

Channel-group listing: Group: 10

Protocol: LACP

SW-2#sh etherchannel protocol

Channel-group listing: Group: 10

Protocol: LACP

SW-1#sh etherchannel summary

Flags: D - down P - bundled in port-channel
 I - stand-alone s - suspended
 H - Hot-standby (LACP only)
 R - Layer3 S - Layer2
 U - in use N - not in use, no aggregation
 f - failed to allocate aggregator
 M - not in use, minimum links not met
 m - not in use, port not aggregated due to minimum links not met
 u - unsuitable for bundling
 w - waiting to be aggregated
 d - default port
 A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators: 1

Group	Port-channel	Protocol	Ports		
10	Po10(SU)	LACP	Gi0/0(P)	Gi0/1(P)	Gi0/2(P)

SW-2#sh etherchannel summary

Flags: D - down P - bundled in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use N - not in use, no aggregation
f - failed to allocate aggregator
M - not in use, minimum links not met
m - not in use, port not aggregated due to minimum links not met
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators: 1

Group	Port-channel	Protocol	Ports		
10	Po10(SU)	LACP	Gi0/0(P)	Gi0/1(P)	Gi0/2(P)

SW-1#sh etherchannel port-channel

Channel-group listing:

Group: 10

Port-channels in the group:

Port-channel: Po10 (Primary Aggregator)

Age of the Port-channel = 0d:00h:12m:57s
Logical slot/port = 16/0 Number of ports = 3
HotStandBy port = null
Port state = Port-channel Ag-Inuse
Protocol = LACP
Port security = Disabled

Ports in the Port-channel:

Index	Load	Port	EC state	No of bits
0	00	Gi0/0	Active	0


```

0 00 Gi0/1 Active 0
0 00 Gi0/2 Active 0

```

Time since last port bundled: 0d:00h:06m:18s Gi0/2
 Time since last port Un-bundled: 0d:00h:06m:25s Gi0/2

SW-2#sh etherchannel port-channel

Channel-group listing:

Group: 10

Port-channels in the group:

Port-channel: Po10 (Primary Aggregator)

Age of the Port-channel = 0d:00h:11m:23s
 Logical slot/port = 16/0 **Number of ports = 3**
 HotStandBy port = null
 Port state = Port-channel Ag-Inuse
Protocol = LACP
 Port security = Disabled

Ports in the Port-channel:

Index	Load	Port	EC state	No of bits
0	00	Gi0/0	Active	0
0	00	Gi0/1	Active	0
0	00	Gi0/2	Active	0

Time since last port bundled: 0d:00h:07m:42s Gi0/2
 Time since last port Un-bundled: 0d:00h:07m:49s Gi0/2

SW-1#sh etherchannel load-balance

EtherChannel Load-Balancing Configuration: src-dst-ip

EtherChannel Load-Balancing Addresses Used Per-Protocol:

Non-IP: Source XOR Destination MAC address

IPv4: Source XOR Destination IP address

IPv6: Source XOR Destination IP address

SW-2#sh etherchannel load-balance

EtherChannel Load-Balancing Configuration: src-dst-ip

EtherChannel Load-Balancing Addresses Used Per-Protocol:

Non-IP: Source XOR Destination MAC address

IPv4: Source XOR Destination IP address

IPv6: Source XOR Destination IP address

Task-4: Shut down the int Gi0/0 (member of EtherChannel) on SW-1

```
SW-1#config t
SW-1(config)#int Gi0/0
SW-1(config-if)#shut
SW-1(config-if)#exit
SW-1(config)#exit
SW-1#
```

✓ Verification & Testing:**SW-1#sh int status | exclude notconnect**

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		disabled	1	auto	auto	RJ45
Gi0/1		connected	trunk	a-full	auto	RJ45
Gi0/2		connected	trunk	a-full	auto	RJ45
Po10		connected	trunk	a-full	auto	

SW-1#sh etherchannel summary

Flags: **D - down** **P - bundled in port-channel**
 I - stand-alone s - suspended
 H - Hot-standby (LACP only)
 R - Layer3 **S - Layer2**
U - in use N - not in use, no aggregation
 f - failed to allocate aggregator
 M - not in use, minimum links not met
 m - not in use, port not aggregated due to minimum links not met
 u - unsuitable for bundling
 w - waiting to be aggregated
 d - default port
 A - formed by Auto LAG

Number of channel-groups in use: 1
 Number of aggregators: 1

Group	Port-channel	Protocol	Ports
10	Po10(SU)	LACP	Gi0/0(D) Gi0/1(P) Gi0/2(P)

SW-2#sh int status | exclude notconnect

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		suspended	trunk	a-full	auto	RJ45
Gi0/1		connected	trunk	a-full	auto	RJ45
Gi0/2		connected	trunk	a-full	auto	RJ45
Po10		connected	trunk	a-full	auto	

SW-2#sh etherchannel summary

Flags: **D - down** **P - bundled in port-channel**
 I - stand-alone **s - suspended**

H - Hot-standby (LACP only)

R - Layer3 **S - Layer2**

U - in use N - not in use, no aggregation

f - failed to allocate aggregator

M - not in use, minimum links not met

m - not in use, port not aggregated due to minimum links not met

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

A - formed by Auto LAG

Number of channel-groups in use: 1

Number of aggregators: 1

Group Port-channel Protocol Ports

-----+-----+-----+-----+-----+-----
10 Po10(SU) LACP Gi0/0(s) Gi0/1(P) Gi0/2(P)

Important Commands:

```
sh spanning-tree  
sh interface trunk  
sh int status | exclude notconnect  
sh lacp neighbor  
sh etherchannel protocol  
sh etherchannel summary
```

Ratan